Database Keys aur Relationships

Beginner-friendly Roman Urdu guide | Inventory, Purchases, Sales aur Warehouses ke examples

Learning order: Pehle table/row samjhein, phir Primary Key, Foreign Key aur Unique; us ke baad One-to-Many, One-to-One, Many-to-Many aur apne inventory system ki relations.

1. Database ka basic concept

Table = data rakhne ki jagah, jaise products. Row = ek record, jaise ek desk. Column = us record ki property, jaise id, name, sku aur category_id.

products.id	products.name	products.sku	products.category_id
1	Office Desk	DESK-001	10
2	Chair	CHAIR-002	10
3	Laptop	LAP-003	20

Is example mein har product ki apni ID hai. category_id batata hai ke product kis category ka hai.

2. PK, FK aur Unique - 3 important rules

Rule	Aasaan matlab	Example
Primary Key (PK)	Har row ki alag, non-null pehchan.	products.id = 1
Foreign Key (FK)	Child table ka column parent table ki existing key ko refer karta hai.	products.category_id -> categories.id
Unique	Ek column ya columns ke combination mein duplicate value rokta hai.	products.sku = DESK-001

School example: Student ka roll number = PK. Us ki class_id = FK (classes table se relation). Student ka email = Unique ho sakta hai.

Primary aur Foreign Key ka mini diagram

categories [id: 10, name: Furniture] 1 ---- many products [id: 1, category_id: 10]

Agar products.category_id = 999 ho aur categories.id = 999 exist na kare, to active FK constraint aam tor par insert/update ko reject karegi. FK ke baghair aisi galat link ban sakti hai.

3. Database relations ki types

Type	Simple explanation	Real example
One-to-Many (hasMany)	Ek parent ki multiple child rows; har child ek parent se linked.	1 supplier -> many purchases
One-to-One (hasOne)	Ek parent ki zero ya ek child row, unique FK ke saath.	1 sales_return -> 0/1 completion
Many-to-Many (belongsToMany)	Dono sides par multiple records; pivot/join table lagti hai.	users <-> roles via model_has_roles

Relationship diagram - 3 shapes

1 : N categories (1) --------> products (many)

1 : 0..1 sales_returns (1) --------> sales_return_completions (0 ya 1)

N : N users (many) --- model_has_roles --- roles (many)

Laravel Eloquent: parent side par aksar hasMany(), child side par belongsTo(), one-to-one par hasOne(), aur many-to-many par belongsToMany() use hota hai.

4. Aapke inventory system ki main relations

A. Access control

Relation	Type	Samajhne ka tareeqa
users <-> roles	Many-to-Many	Ek user ke multiple roles; role ke multiple users. Pivot: model_has_roles
roles <-> permissions	Many-to-Many	Ek role ko multiple permissions; permission multiple roles ko. Pivot: role_has_permissions
users <-> warehouses	Many-to-Many	User ko multiple warehouses assign ho sakte hain. Pivot: user_warehouse

B. Catalog aur inventory

Relation	Type / meaning
categories -> products	One-to-Many: category ke bohat se products
products + warehouses -> inventory_balances	Har product-warehouse pair ka sirf ek current balance (composite unique constraint)
products / warehouses -> stock_movements	Har stock change ki multiple history rows; movements product aur warehouse ko refer karti hain

C. Purchase process

suppliers (1) -> purchases (many) -> purchase_items (many) -> products (1 per line)

purchases (1) -> purchase_receipts (many) -> purchase_receipt_items (many) -> purchase_items (1 per receipt line)

D. Sales aur returns

customers -> sales_orders -> sales_order_items -> products

sales_orders -> sales_returns -> sales_return_items

sales_returns (1) -> sales_return_completions (0 ya 1)

E. Transfers aur adjustments

warehouse_transfers ki source_warehouse_id aur destination_warehouse_id, dono warehouses.id ko refer karti hain. Transfer ke multiple transfer_items aur receipts ho sakte hain. stock_adjustments product, warehouse aur creator user se linked hoti hain.

5. Foreign Keys - kaun kis table se linked hai?

Child table	Foreign key columns	Parent table(s)
products	category_id	categories
inventory_balances	product_id, warehouse_id	products, warehouses
stock_movements	product_id, warehouse_id, user_id	products, warehouses, users
purchases	supplier_id, warehouse_id, created_by, approved_by	suppliers, warehouses, users
purchase_items	purchase_id, product_id	purchases, products
purchase_receipt_items	purchase_receipt_id, purchase_item_id	purchase_receipts, purchase_items
sales_orders	customer_id, warehouse_id	customers, warehouses
sales_return_completions	sales_return_id	sales_returns
warehouse_transfers	source_warehouse_id, destination_warehouse_id	warehouses (dono)
stock_adjustments	product_id, warehouse_id, created_by	products, warehouses, users

6. Unique rules - duplicate data kahan rukta hai?

Table	Unique field(s)	Faida
products	sku	Ek SKU sirf ek product ka
inventory_balances	product_id + warehouse_id	Per product/warehouse sirf ek balance
user_warehouse	user_id + warehouse_id	Same assignment duplicate nahi
stock_alert_states	product_id + warehouse_id	Ek pair ki ek alert state
purchase_receipts	idempotency_key	Repeat receive request se duplicate processing roki ja sakti hai
warehouse_transfer_receipts	idempotency_key	Repeat transfer receipt protection
sales_return_completions	idempotency_key	Repeat return completion protection
purchase / sales documents	document number (as configured)	Repeated document number roka jata hai
purchase_items	purchase_id + product_id	Ek purchase mein product ek line
sales_order_items	sales_order_id + product_id	Ek order mein product ek line

Important: UNIQUE constraint duplicate idempotency_key ko reject kar sakta hai, lekin purana response wapas dena application logic ka kaam hai. Nullable UNIQUE columns ka behavior database ke mutabiq badal sakta hai.

7. Practical example: 10 desks ki purchase

Sochiye supplier se 10 desks ka purchase order bana. Delivery ek hi waqt nahi aayi: pehli baar 6 desks, phir 4 desks.

Table	Rows	Kya save hoga
purchases	1	Ek purchase order
purchase_items	1	Desk product, ordered quantity = 10
purchase_receipts	2	Receipt #1 aur receipt #2
purchase_receipt_items	2	Pehli receipt: 6; doosri: 4
stock_movements	2 (simple case)	+6 aur +4 ki stock history
inventory_balances	1 current row	Us warehouse mein desk quantity net +10 (agar aur koi movement nahi)

Flow: Purchase create -> items add -> receipt #1 (+6) -> receipt #2 (+4) -> current balance update -> movements history save.

8. Polymorphic relationship kya hoti hai?

Kuch links mein type + id save hota hai: misaal ke taur par model_type = Product aur model_id = 5. Isi structure se record different models ko refer kar sakta hai. Ye notifications, personal_access_tokens, model_has_roles aur stock_movements ke polymorphic reference mein mil sakta hai.

Aam relational databases ek type-dependent ID par standard FK constraint seedha nahi laga sakte. Is liye matching model aur ID ki correctness application logic ko bhi check karni padti hai.

9. Interview / revision ke liye short answers

PK? Table ki har row ka unique ID. FK? Doosri table ki row ki reference. Unique? Duplicate values ko rokne ka rule.

hasMany? Ek parent ki multiple children. hasOne? Ek parent ki maximum ek related row. belongsToMany? Pivot table ke through many-to-many.

Inventory balance vs stock movement? Balance abhi kitna stock hai; movement batati hai stock kab aur kyun change hua.

> Yaad rakhne ka shortcut: PK = Main kaun hoon? FK = Main kis se linked hoon? UNIQUE = Kya ye value pehle se maujood hai?